

Chapter 3

Content-based Recommender Systems: State of the Art and Trends

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Abstract Recommender systems have the effect of guiding users in a personalized way to interesting objects in a large space of possible options. *Content-based* recommendation systems try to recommend items similar to those a given user has liked in the past. Indeed, the basic process performed by a content-based recommender consists in matching up the attributes of a user profile in which preferences and interests are stored, with the attributes of a content object (item), in order to recommend to the user new interesting items. This chapter provides an overview of content-based recommender systems, with the aim of imposing a degree of order on the diversity of the different aspects involved in their design and implementation. The first part of the chapter presents the basic concepts and terminology of content-based recommender systems, a high level architecture, and their main advantages and drawbacks. The second part of the chapter provides a review of the state of the art of systems adopted in several application domains, by thoroughly describing both classical and advanced techniques for representing items and user profiles. The most widely adopted techniques for learning user profiles are also presented. The last part of the chapter discusses trends and future research which might lead towards the next generation of systems, by describing the role of User Generated Content as a way for taking into account evolving vocabularies, and the challenge of feeding users with serendipitous recommendations, that is to say surprisingly interesting items that they might not have otherwise discovered.

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3.1 Introduction

The abundance of information available on the Web and in Digital Libraries, in combination with their dynamic and heterogeneous nature, has determined a rapidly increasing difficulty in finding what we want when we need it and in a manner which best meets our requirements.

As a consequence, the role of user modeling and personalized information access is becoming crucial: users need a personalized support in sifting through large amounts of available information, according to their interests and tastes.

Many information sources embody recommender systems as a way of personalizing their content for users [73]. Recommender systems have the effect of guiding users in a personalized way to interesting or useful objects in a large space of possible options [17]. Recommendation algorithms use input about a customer's interests to generate a list of recommended items. At Amazon.com, recommendation algorithms are used to personalize the online store for each customer, for example showing programming titles to a software engineer and baby toys to a new mother [50].

The problem of recommending items has been studied extensively, and two main paradigms have emerged. *Content-based* recommendation systems try to recommend items similar to those a given user has liked in the past, whereas systems designed according to the *collaborative* recommendation paradigm identify users whose preferences are similar to those of the given user and recommend items they have liked [7].

In this chapter, a comprehensive and systematic study of content-based recommender systems is carried out. The intention is twofold:

- to provide an overview of state-of-the-art systems, by highlighting the techniques which revealed the most effective, and the application domains in which they have adopted;
- to present trends and directions for future research which might lead towards the next generation of content-based recommender systems.

The chapter is organized as follows. First, we present the basic concepts and terminology related to content-based recommenders. A classical framework for providing content-based recommendations is described, in order to understand the main components of the architecture, the process for producing recommendations and the advantages and drawbacks of using this kind of recommendation technique. Section 3.3 provides a thorough review of the state of the art of content-based systems, by providing details about the classical and advanced techniques for representing items to be recommended, and the methods for learning user profiles. Section 3.4 presents trends and future research in the field of content-based recommender systems, while conclusions are drawn in Section 3.5.